

IT Helpdesk Home Lab

Days 11–17

Windows Server 2022 & Active Directory Home Lab

Domain: alexlang-helpdesk.com

OS: Windows Server 2022 + Windows 11 Pro

Hypervisor: VirtualBox

Coverage

Day 11 — DHCP Server Setup

Day 12 — DNS Troubleshooting

Day 13 — PowerShell Automation

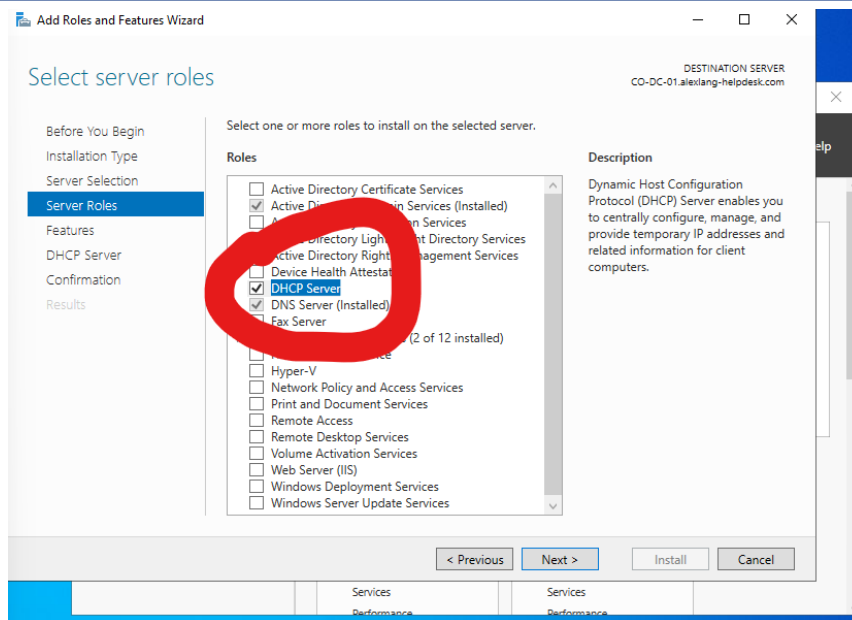
Day 14 — osTicket Ticketing System

Day 15 — NTFS Permissions & Security

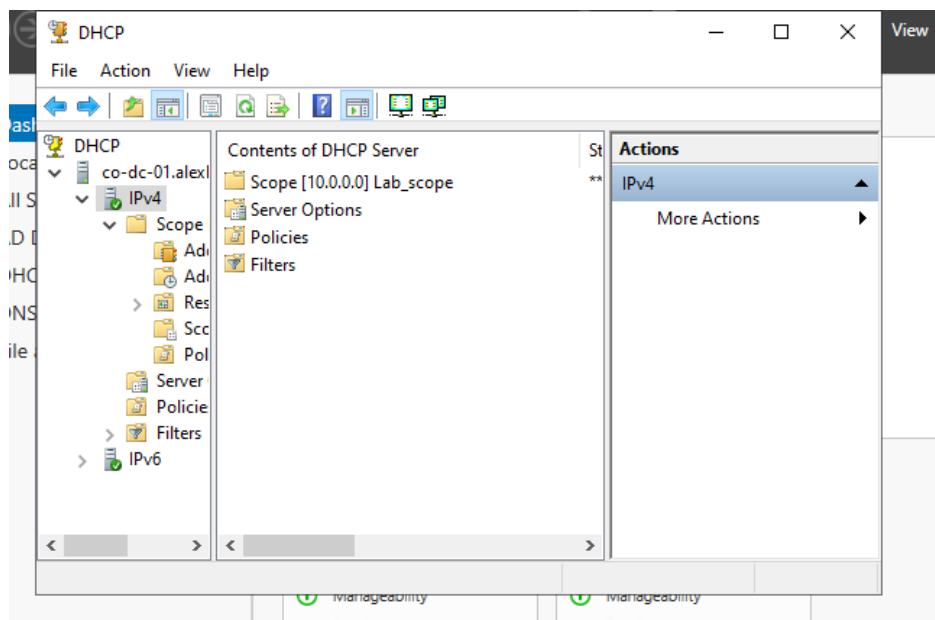
Day 16 — Software Deployment via Group Policy

Day 17 — Backup & Recovery

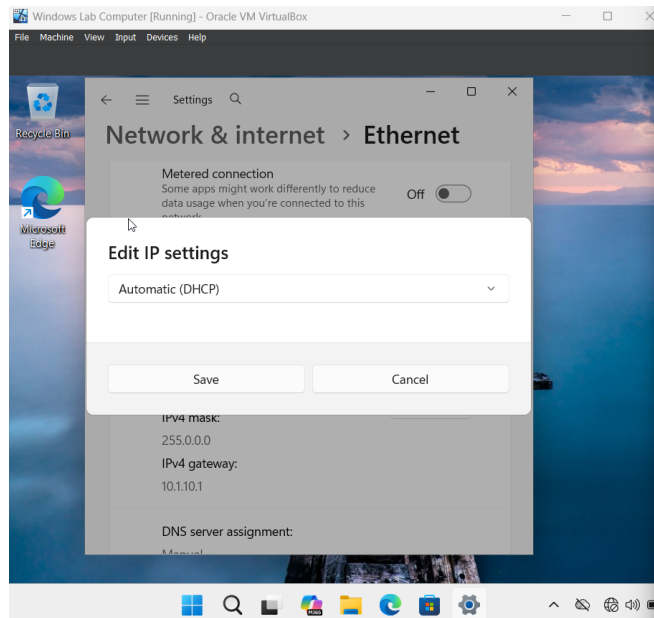
Day 11 Setting up DHCP Server



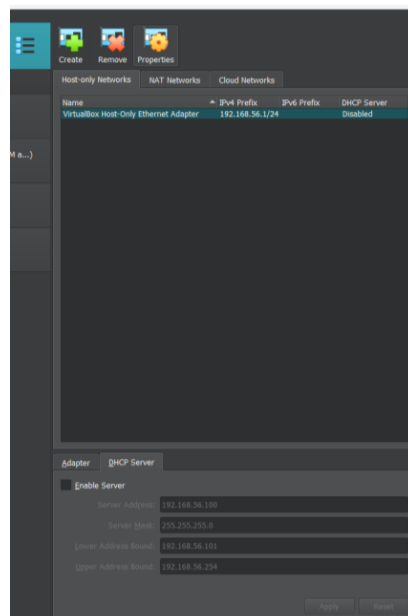
Installing DHCP Server inside of the add roles and features wizard



DHCP custom scope setup, using a custom IP address range, excludes 10.1.10.2 which is my server's static IP, added a lease of 10 days, and used 10.1.10.1 as my default gateway (which is my server's static IP).

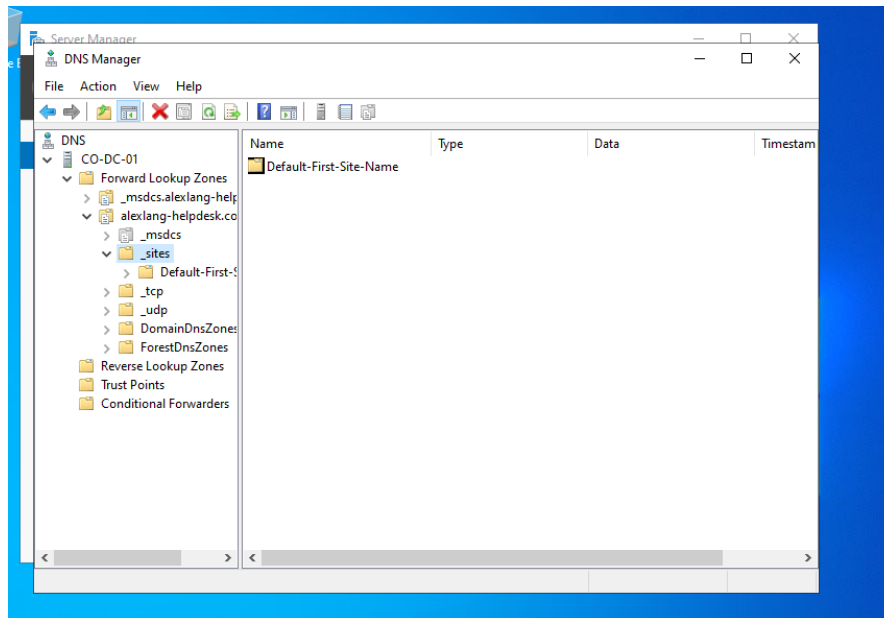


From here, I switched over to my lab computer, went into Network/Internet settings, and changed IP settings to use DHCP instead of static IP assignment.

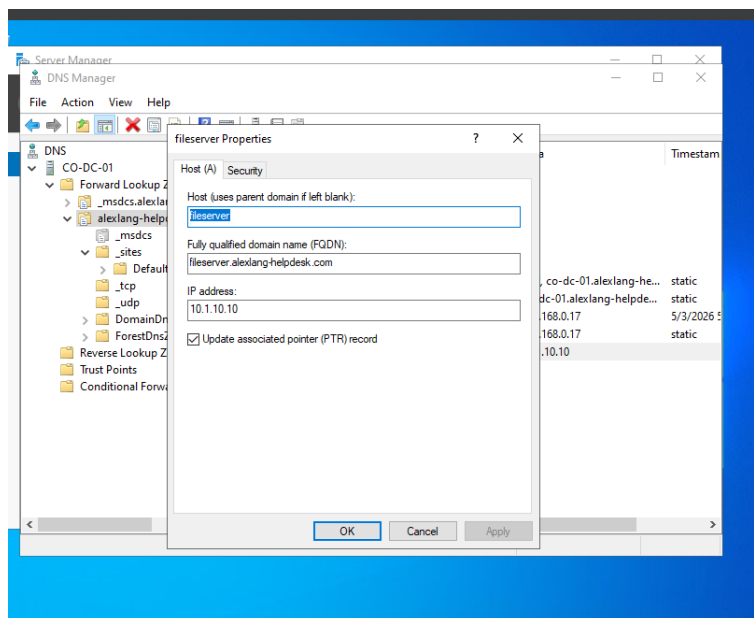


Went into the hypervisor to disable the DHCP Server since we just created one

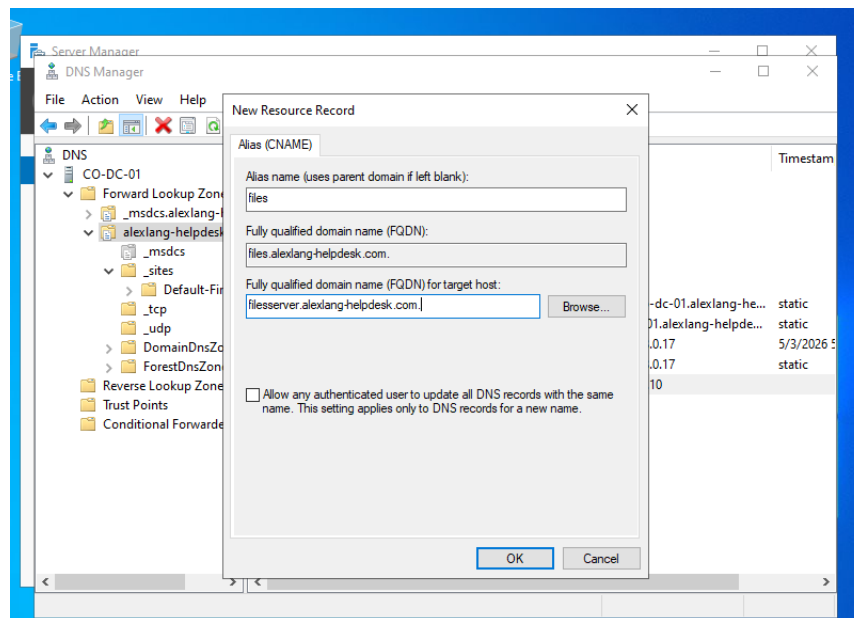
Day 12 DNS Troubleshooting



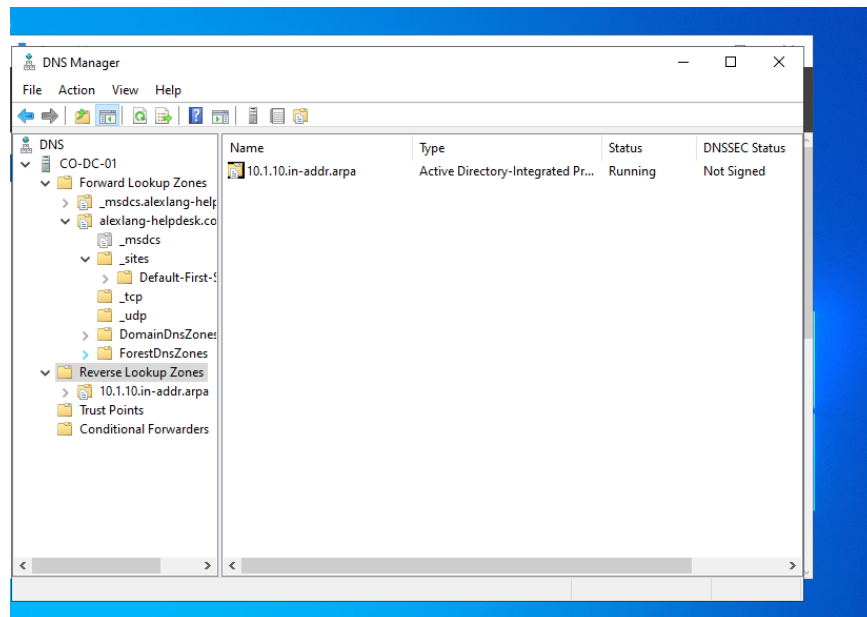
Took some time to examine all the DNS records that were created automatically by AD. Seeing these SOA, NS, A, and SRV records was really refreshing because I learned what each one does while studying for CompTIA's A+.



Created an A record

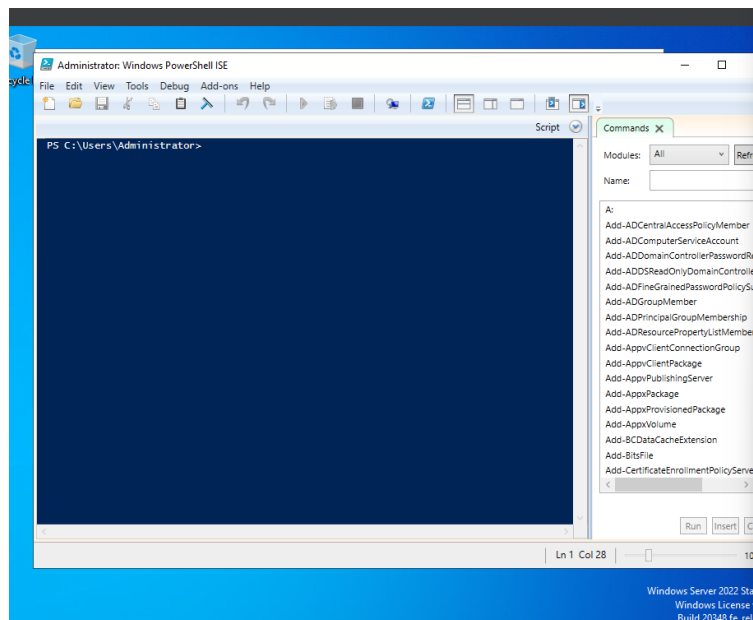


Created a CNAME record also known as an alias for the server just created

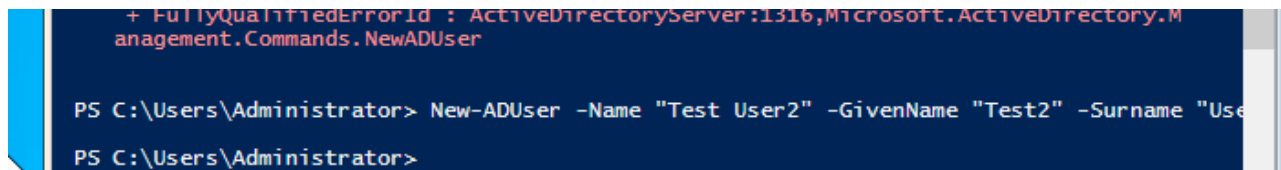


Created a reverse lookup zone

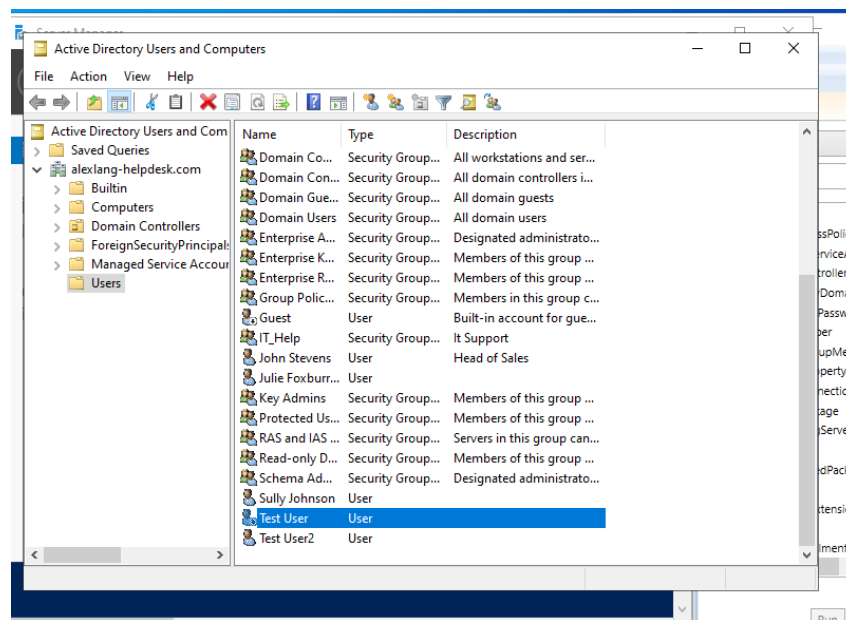
Day 13 PowerShell ISE



Going to write various shell scripts to automate tasks inside of AD



Automated the process of creating user accounts using AD specifications. I did use AI to help me write these scripts for time purposes, and to help me get a better understanding of PowerShell ISE documentation.



Test users 1 and 2 were successfully created

```

Administrator: Windows PowerShell ISE
File Edit View Tools Debug Add-ons Help

createUser.ps1 X
1 $users = @(
2   @{Name="Amy Carter"; Sam="acarter"},
3   @{Name="Brian Lopez"; Sam="blopez"},
4   @{Name="Dana Kim"; Sam="dkim"}
5 )
6
7 foreach ($u in $users) {
8   New-ADUser `
9     -Name $u.Name `
10    -SamAccountName $u.Sam `
11    -UserPrincipalName "$($u.Sam)@alexlang-helpdesk.com" `
12    -Path "CN=Users,DC=alexlang-helpdesk,DC=com" `
13    -AccountPassword (ConvertTo-SecureString "Welcome1!" -AsPlainText -Force) `
14    -Enabled $true
15   Write-Host "Created: $($u.Name)"
16 }

PS C:\Users\Administrator>

```

Created a .ps1 file so that the creation of user accounts can be automated a lot easier

```

Administrator: Windows PowerShell ISE
File Edit View Tools Debug Add-ons Help

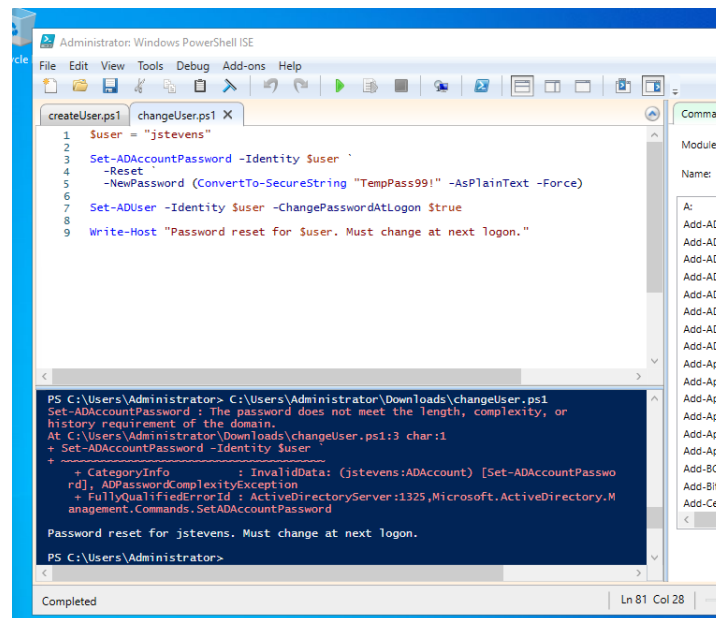
createUser.ps1 X
1 $users = @(
2   @{Name="Amy Carter"; Sam="acarter"},
3   @{Name="Brian Lopez"; Sam="blopez"},
4   @{Name="Dana Kim"; Sam="dkim"}
5 )
6
7 foreach ($u in $users) {
8   New-ADUser `
9     -Name $u.Name `
10    -SamAccountName $u.Sam `
11    -UserPrincipalName "$($u.Sam)@alexlang-helpdesk.com" `
12    -Path "CN=Users,DC=alexlang-helpdesk,DC=com" `
13    -AccountPassword (ConvertTo-SecureString "Welcome1!" -AsPlainText -Force) `
14    -Enabled $true
15   Write-Host "Created: $($u.Name)"
16 }

management.Commands.NewADUser
Created: Dana Kim

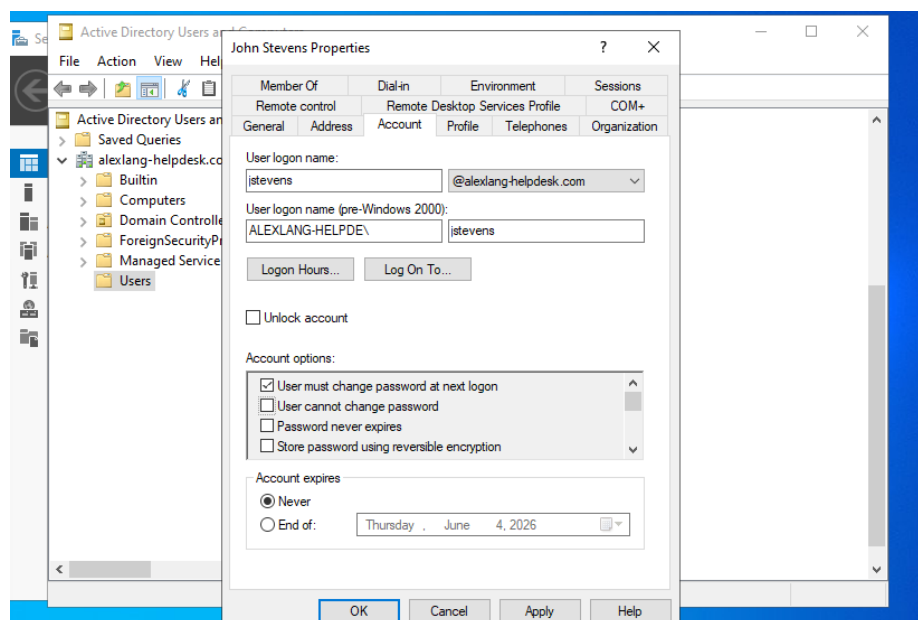
PS C:\Users\Administrator>

```

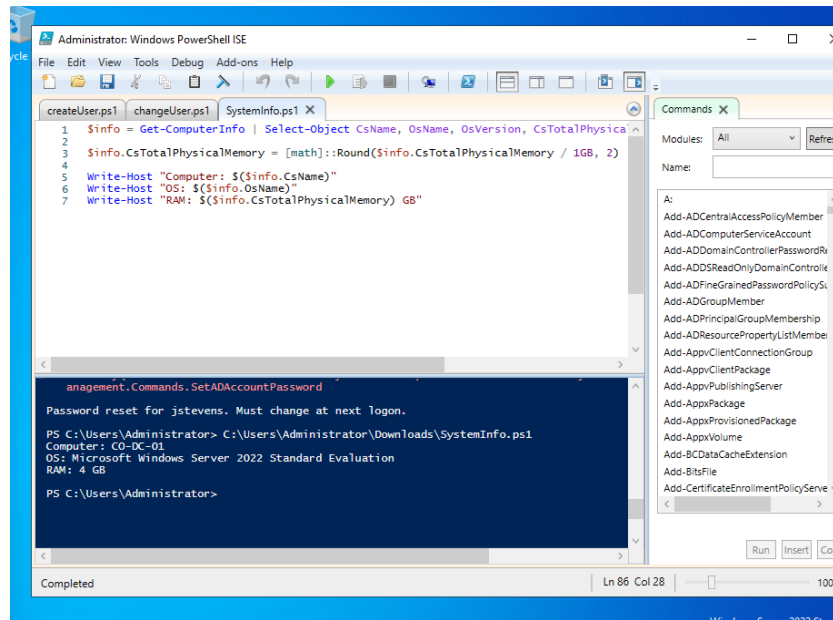
Showing the creation of given user accounts



Wrote a script for password reset as well, now user jstevens must reset their password next time they try to login to their workstation

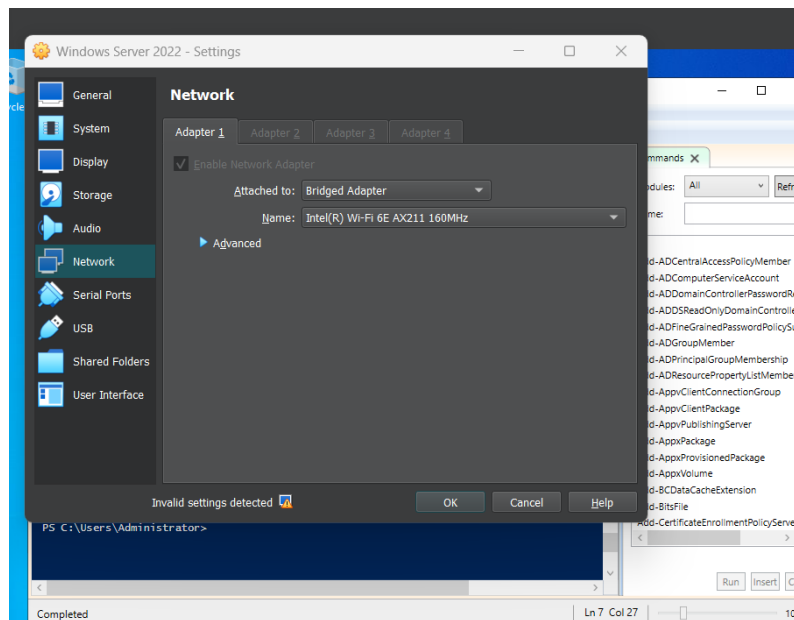


Navigating back to AD Users and Groups, it clearly said that the user must change their login meaning the script ran successfully

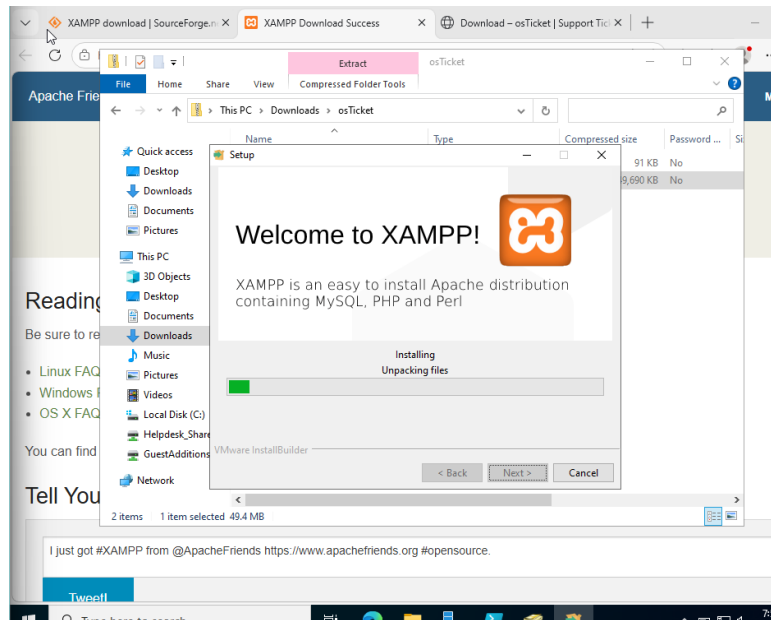


Wrote one more to get some basic info about the system

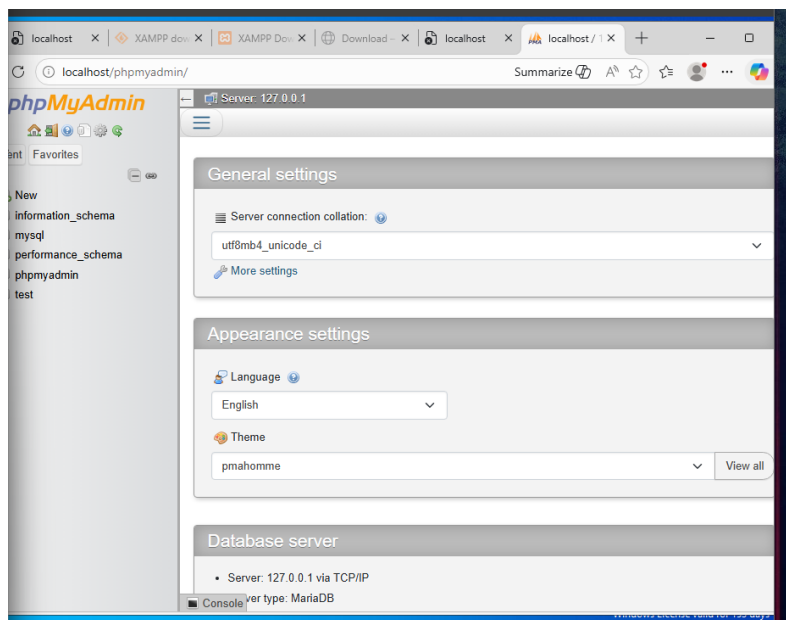
Day 14 Ticketing System with OS Ticket



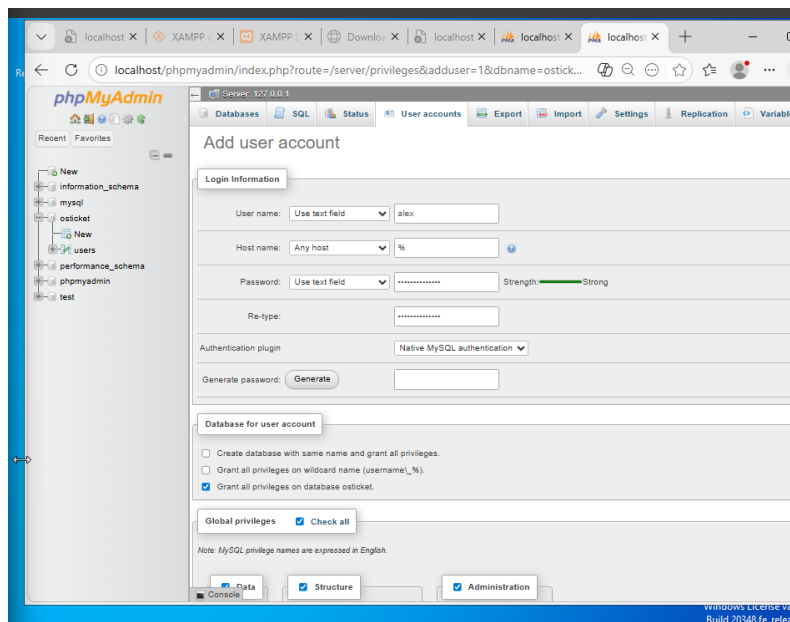
During most of this experiment so far my VM has been on a host-only network connection but I need a network connection to install osTicket so temporarily making it a bridged adapter



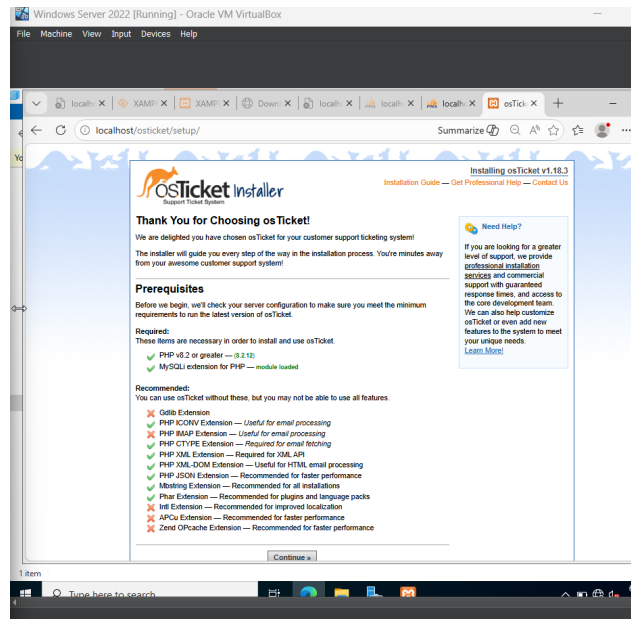
Installed XAMPP(Apache) so I have a local webserver that osTicket can run on and then installed osTicket on the XAMPP server.



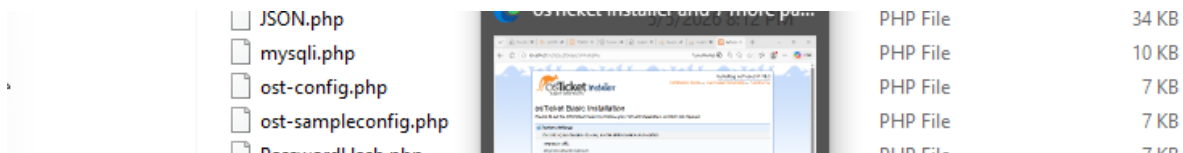
XAMPP Control Panel successfully configured, and phpMyAdmin successfully loaded



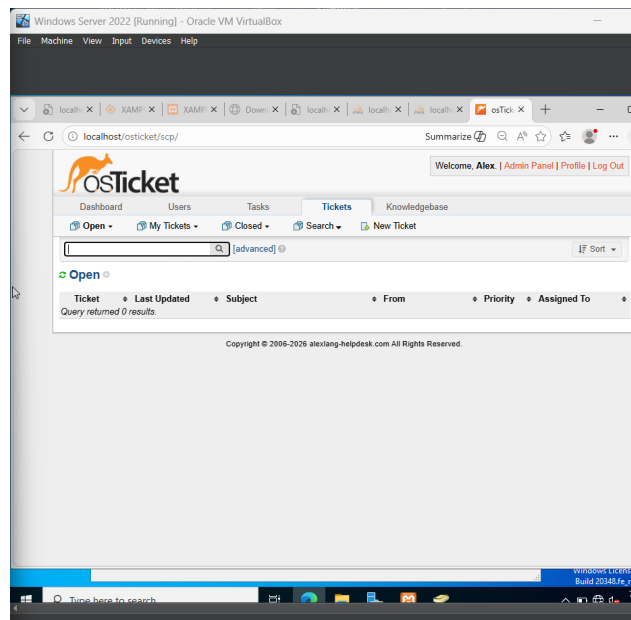
Created a database, added a user account, and gave that account admin permissions



Had to move some files around and rename some folders, but finally got setup configured in the right location



Updated the config file so that the server could find it



OSTicket dashboard

Departments ➕ Add New Department ⚙ More

☐	Name	Status	Type	Agents	Email Address	Manager	Created
☐	IT Support	Active	Public	0	Support <alexlanghelpdesk@gmail.com>		2026-05-06 07:52:53
☐	Maintenance	Active	Public	0	Support <alexlanghelpdesk@gmail.com>		2026-05-06 07:28:07
☐	Network Operations	Active	Public	0	Support <alexlanghelpdesk@gmail.com>		2026-05-06 07:53:10
☐	Sales	Active	Public	0	Support <alexlanghelpdesk@gmail.com>		2026-05-06 07:28:06
☐	Support (Default)	Active	Public	1	Support <alexlanghelpdesk@gmail.com>		2026-05-06 07:28:06

Select: All None Toggle
Page: [1]

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Created 2 new departments, IT Support and Network Operations

Help Topics ➕ Add New Help Topic ⚙ More

Sorting Mode: Alphabetically

☐	Help Topic	Status	Type	Priority	Department	Last Updated	Created
☐	Account Lockout	Active	Public	Normal	Support	05/06/26 08:03:53	05/06/26 08:02:27
☐	Feedback	Active	Public	Low	Support	05/06/26 07:28:08	05/06/26 07:28:08
☐	General Inquiry	Active	Public	Normal	Support	05/06/26 07:28:08	05/06/26 07:28:08
☐	Hardware Issue	Active	Public	Normal	Support	05/06/26 08:06:18	05/06/26 08:06:18
☐	Password Reset	Active	Public	Normal	Support	05/06/26 08:04:12	05/06/26 08:04:12
☐	Report a Problem	Active	Public	Normal	Maintenance	05/06/26 07:28:08	05/06/26 07:28:08
☐	Report a Problem / Access Issue	Active	Public	High	Support	05/06/26 07:28:08	05/06/26 07:28:08

Select: All None Toggle
Page: [1]

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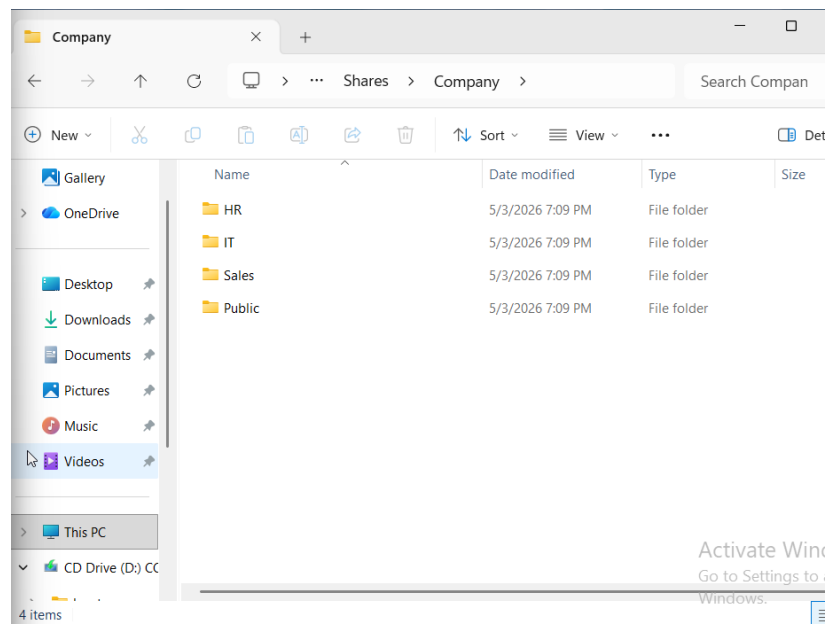
Created some general new help topics

Had a problem with the format of the emails I was using to create tickets, so I had to navigate back to the admin panel and uncheck the setting "Require email verification on ticket status". That also did not work right away, and it would only accept tickets that passed validation. I got stuck here, and couldn't add any further tickets, will try again later

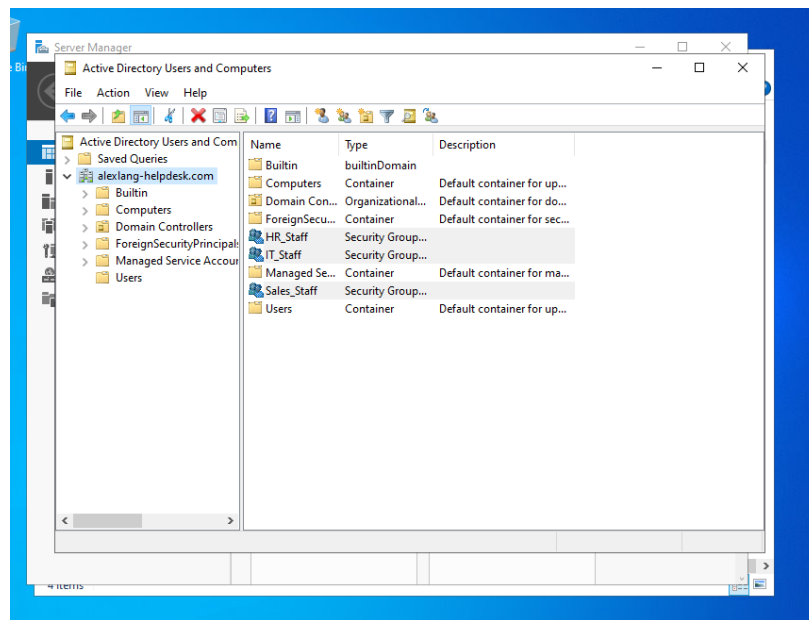
Observations/Problems that I had to troubleshoot through

- I had issues initially setting up the configuration file because the server could not find it (had to do some file configuration management to fix the location issue)
- I also had issues after trying to install osTicket when the server could find the config file and kept getting a permissions error (The configure file admin did not have the proper permissions)

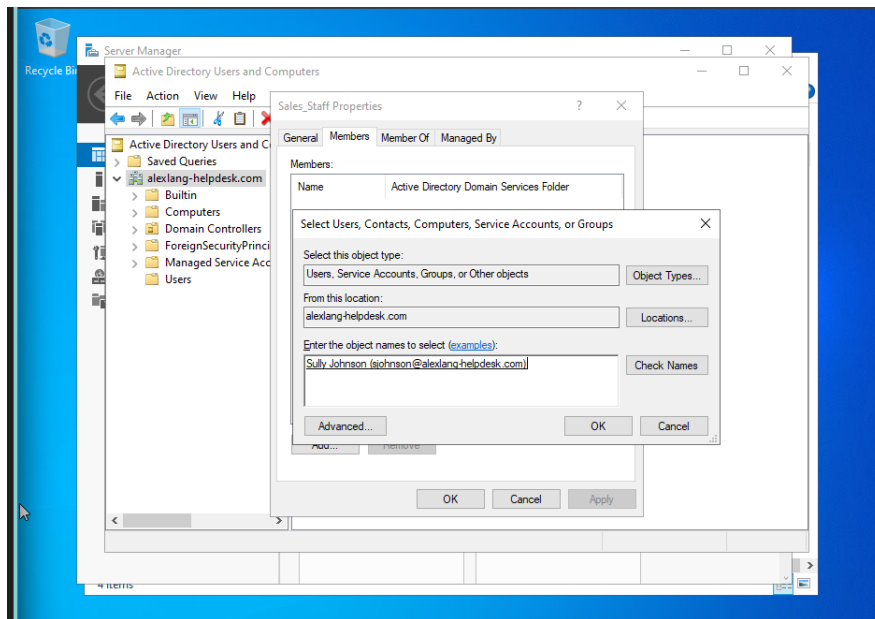
Day 15 NTFS Permissions and Security



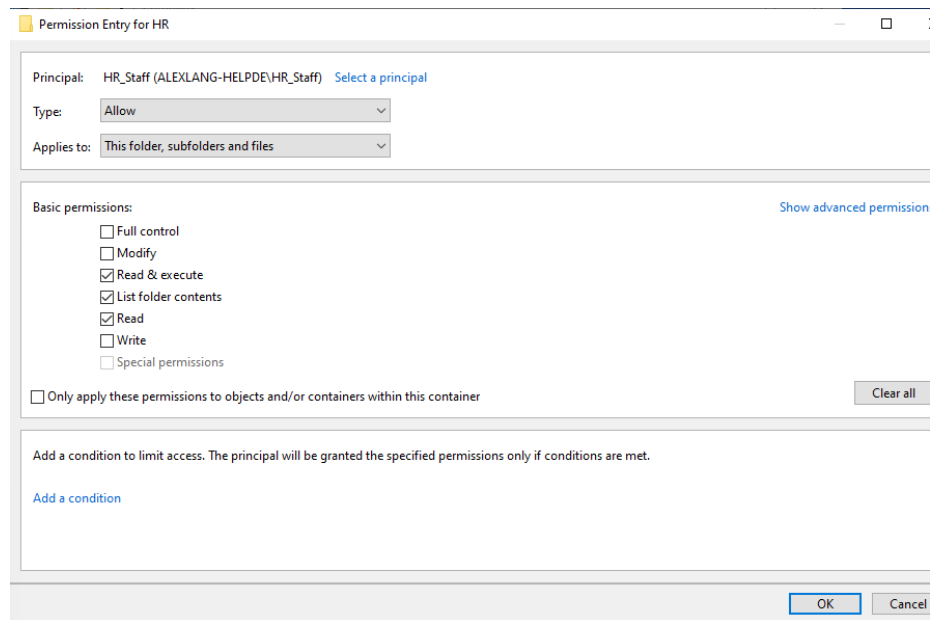
Created a very basic file structure



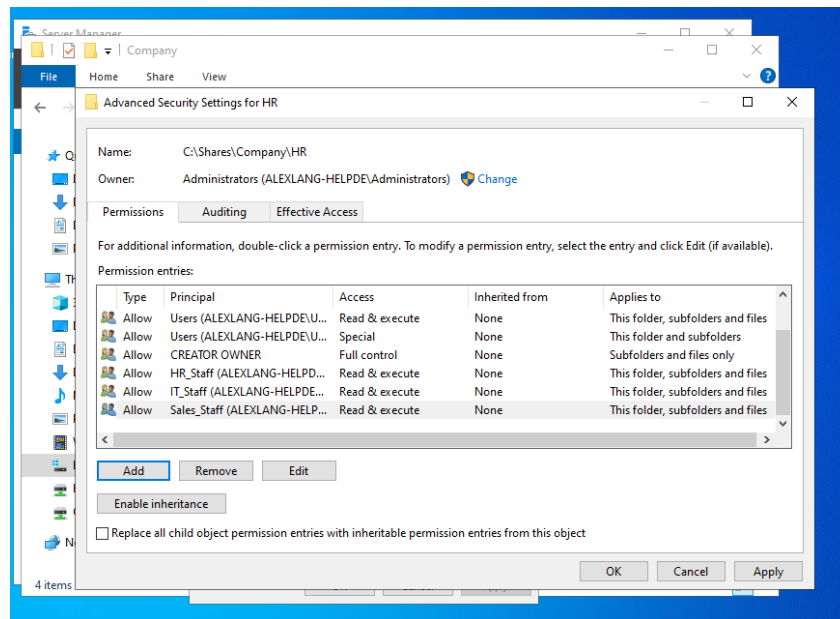
Created 3 new security groups



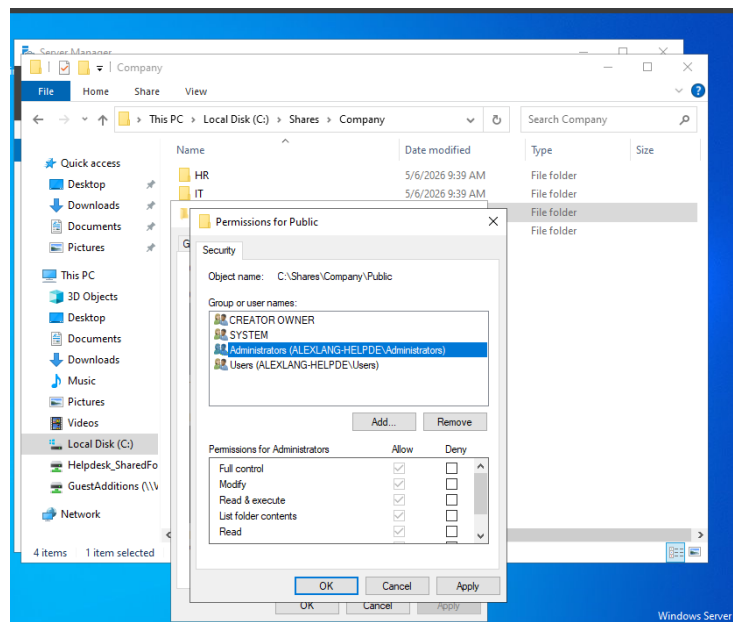
Added appropriate users to these groups that have already been created inside of AD.



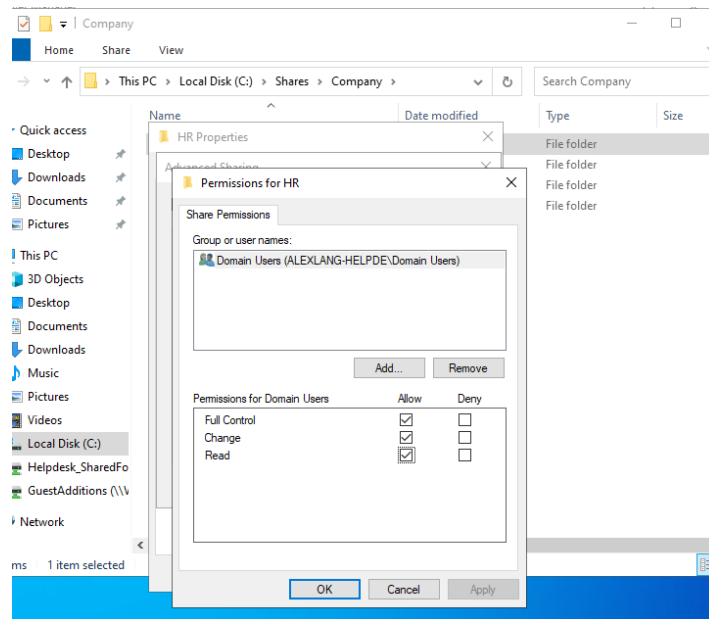
Went into each folder and adjusted inheritance permissions for HR, IT, Sales, and Public



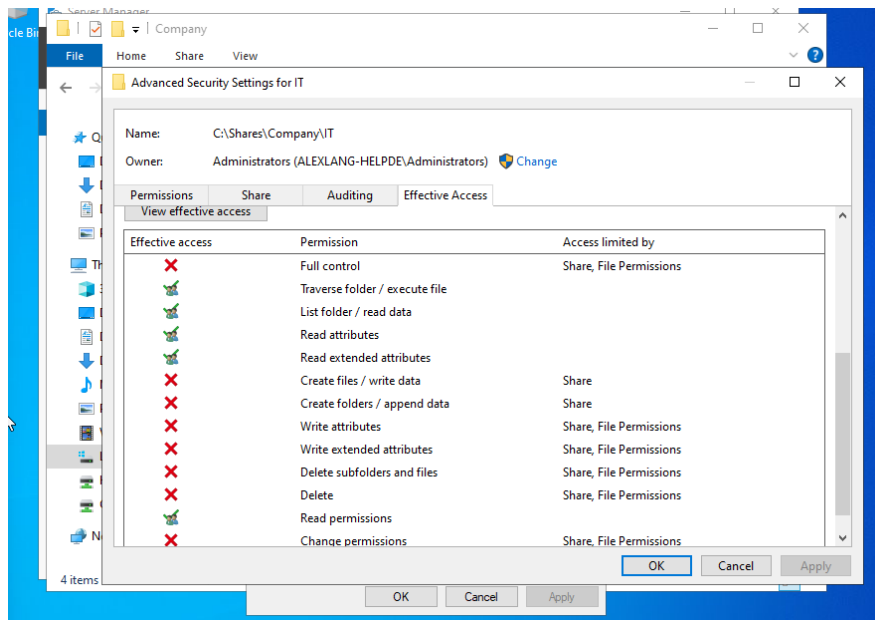
HR, IT, and Sales staff now have read and execute permissions via NTFS permissions



The public folder is supposed to be accessible by everyone, so I made sure everyone has read permissions to it

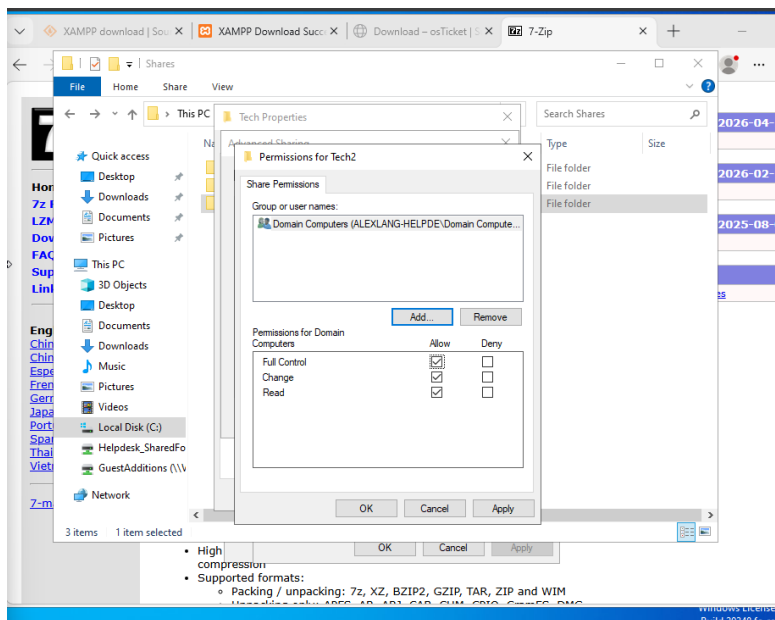


Gave domain users read permissions, and domain admins full control

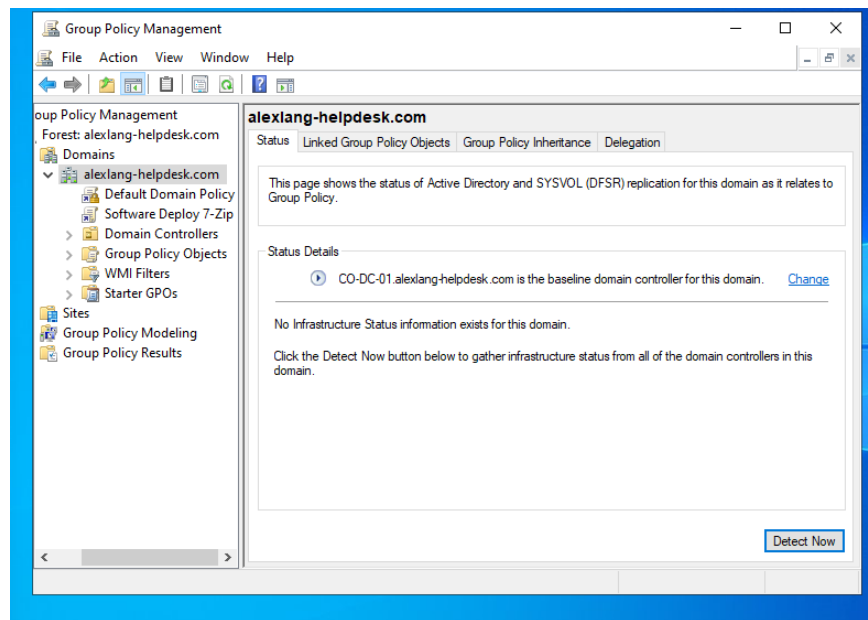


Used effective access control to view all security settings of multiple users to get a full overview of permissions

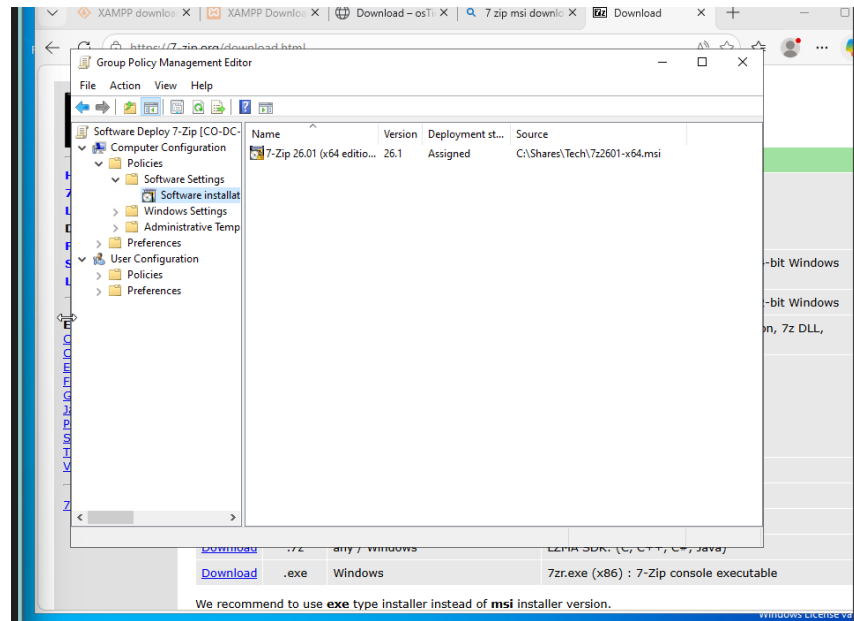
Day 16 Software Deployment through Group Policy



Downloaded 7-Zip on Server controller, and needed to give folder permissions to all domain computers



Created a new group policy object inside of alexlang-helpdesk.com

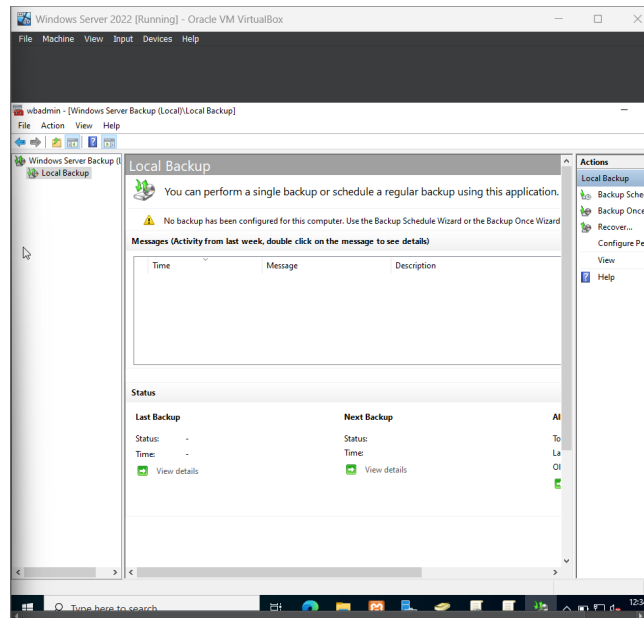


Package successfully added, ran gpupdate on windows machine, restarted vm, and it auto installed it on that machine

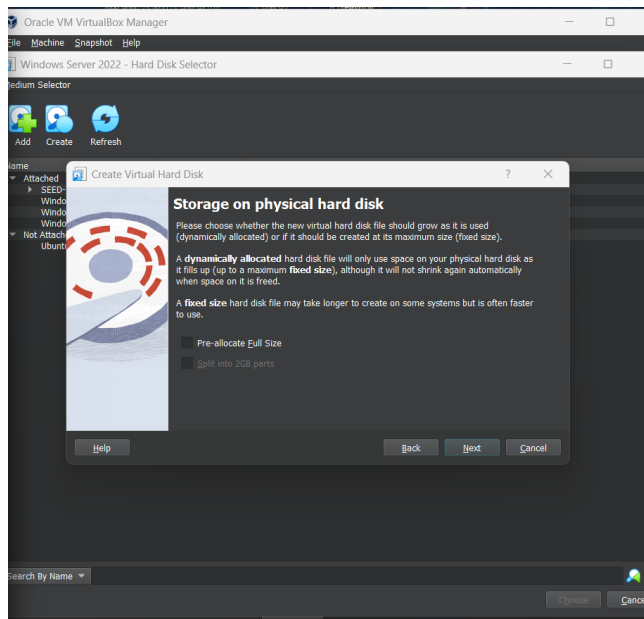
Observations/Problems that I had to troubleshoot through

- I initially installed the wrong version of 7-ZIP, so I had to go back and install the .msi version instead of the .exe version

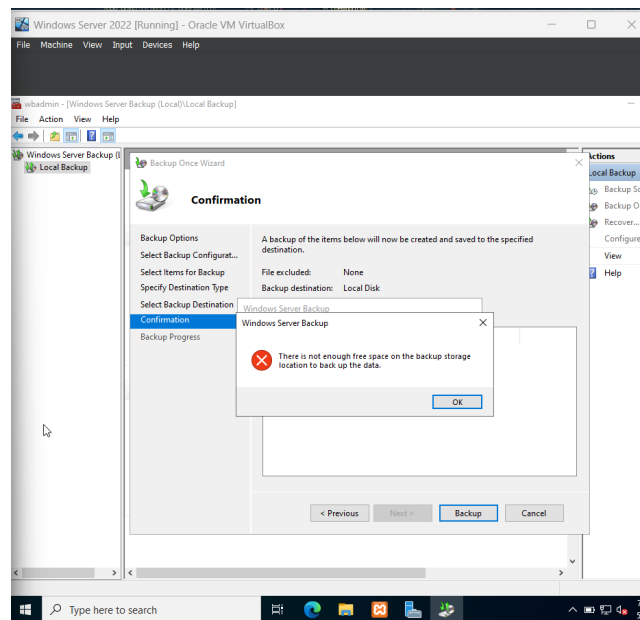
Day 17 Backup and Recovery



Installed Windows Server Backup from add roles/features, and opened it



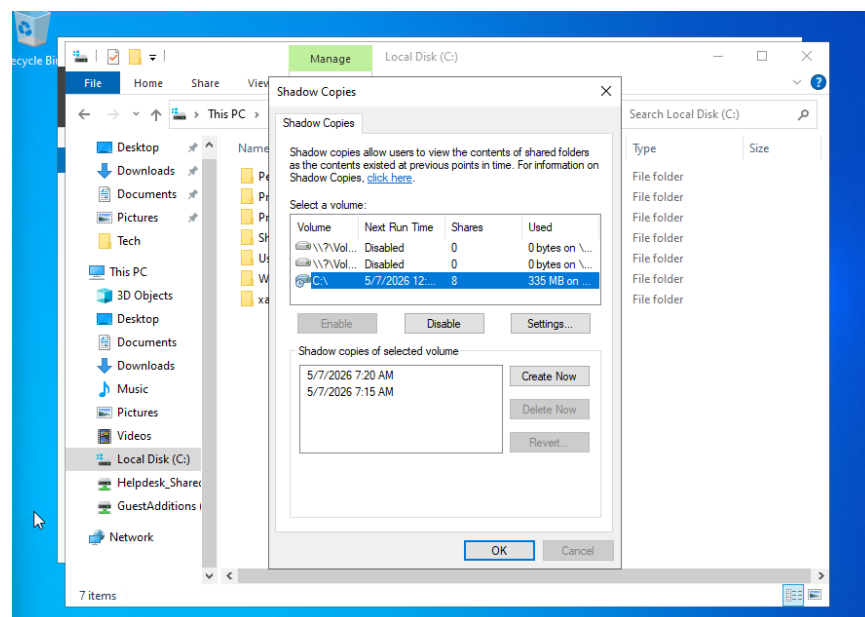
When trying to initially create a manual backup, my VM did not have enough space, so I created a new VHD within the hypervisor.



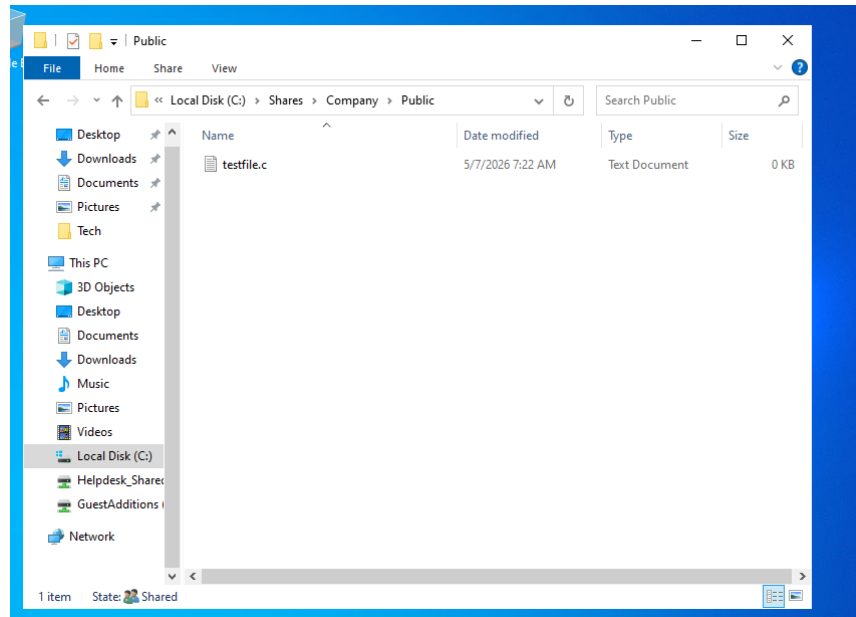
After this troubleshooting, I was still unable to make a successful manual backup. For time purposes I decided just to skip this part and to continue forward with the lab.

Time	Message	Description
 5/6/2026 12:54 PM	Backup	Failed
 5/6/2026 12:51 PM	Backup	Failed
 5/6/2026 12:44 PM	Backup	Failed

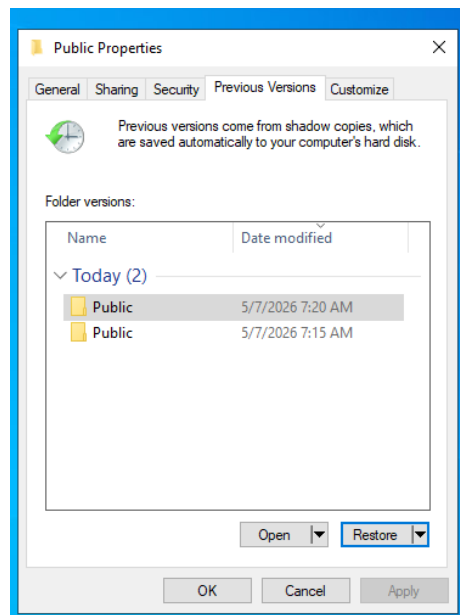
I did however make several attempts to make the backup, backing it up to different storage devices



Setup automatic shadow copy backups of the C drive for each day at 2pm, and also created an immediate copy.



Created a sample file inside of C:\Shares\Company\Public so I can simulate a real world important file becoming deleted

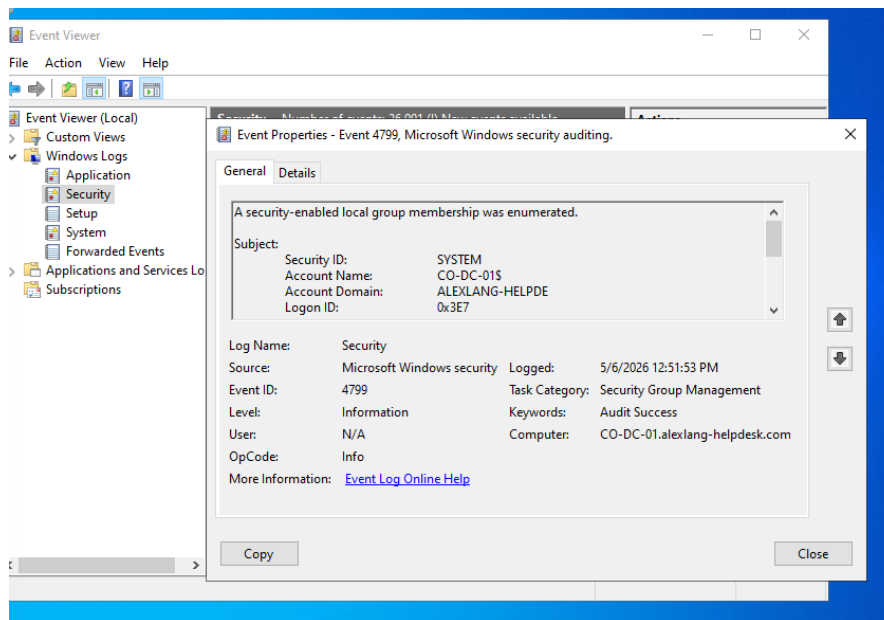


Went into the folder's previous version properties and found the latest version of the folder before testfile.c was deleted.

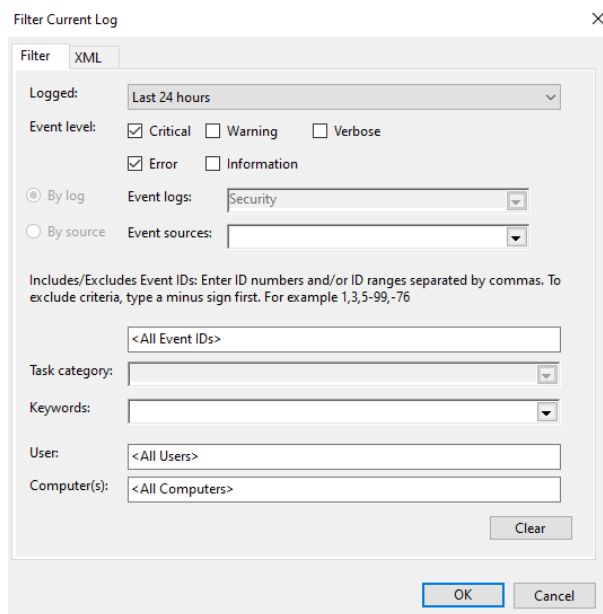
Observations/Problems that I had to troubleshoot through

- I definitely could've figured out the manual backup problem but with limited time to work on this project, I just decided to bypass it

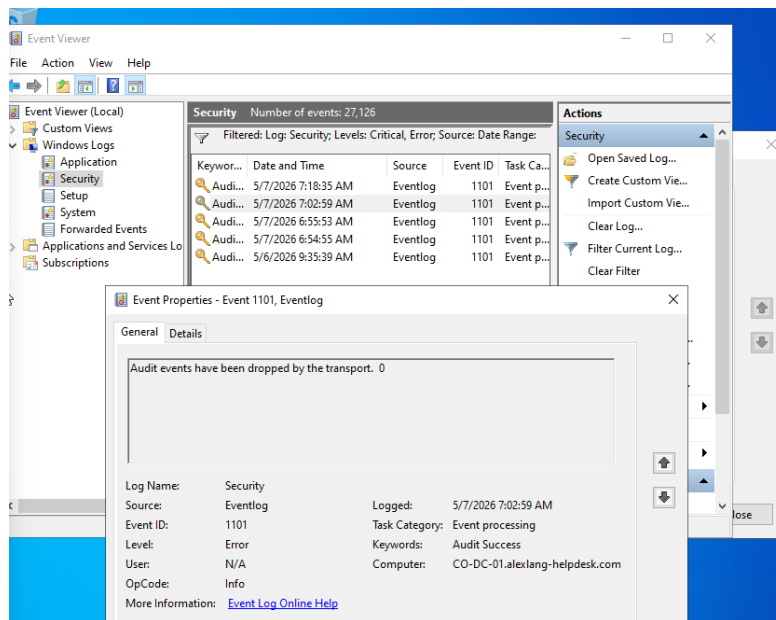
Day 18 Event Viewer and Log Analysis



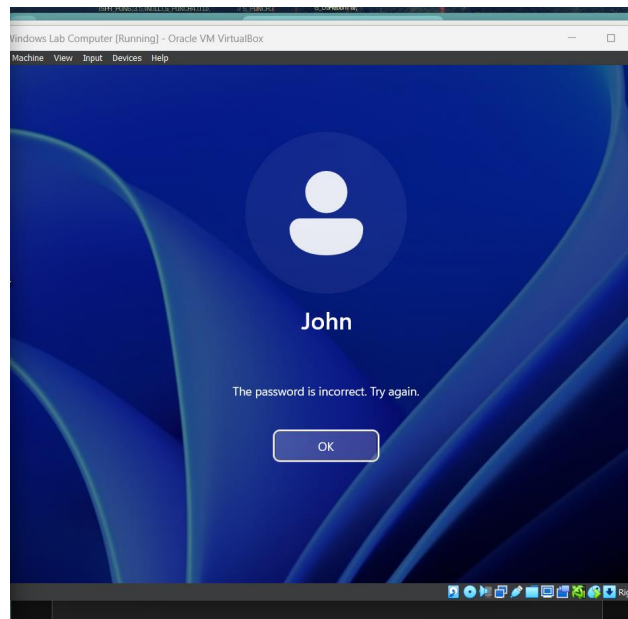
Spent some time exploring Event Viewer, specifically the Windows Logs. It's so cool how often Windows just automatically runs security audits sometimes even several times a second.



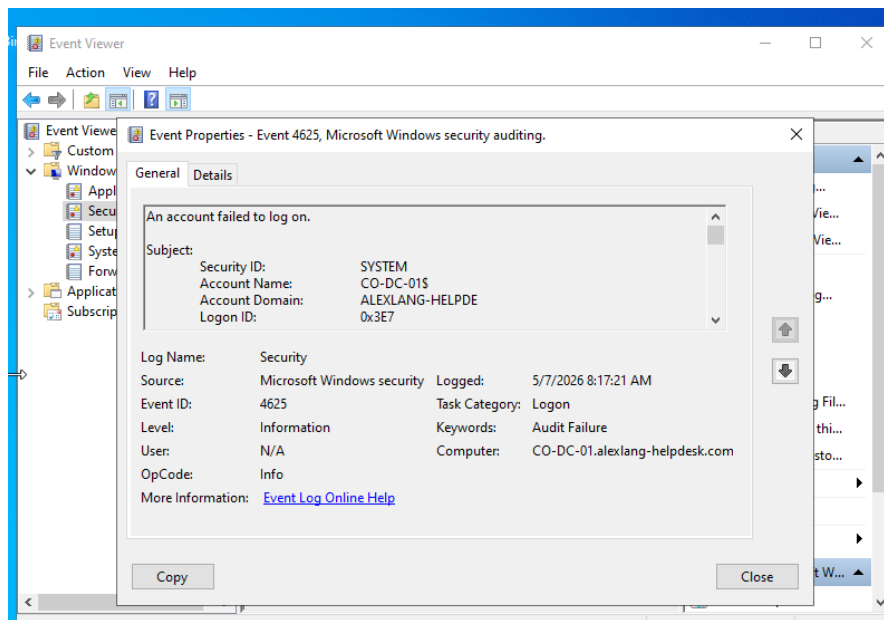
Filtering for only critical/error security logs within the last 24 hours



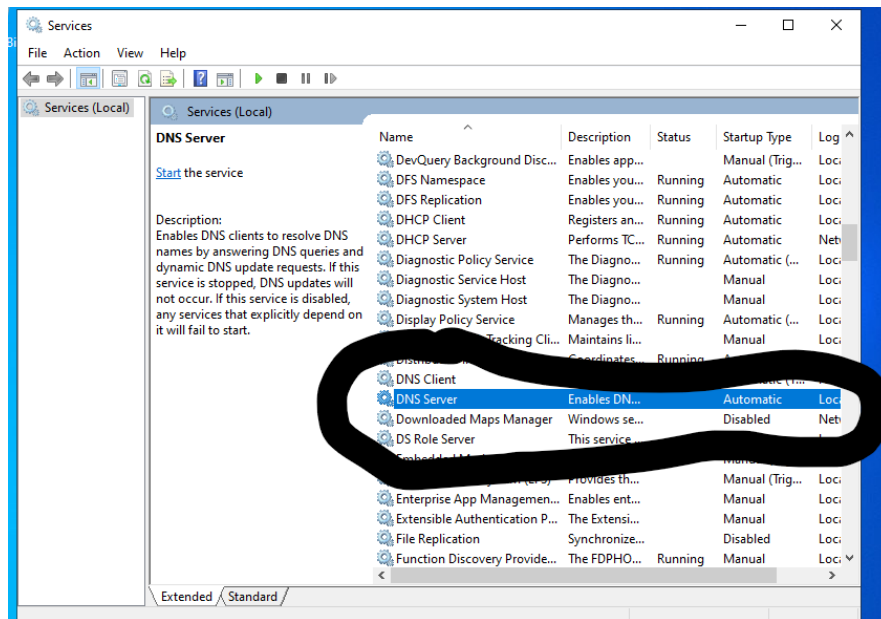
All security logs with this filter were all pretty much the same just spread across a couple hours



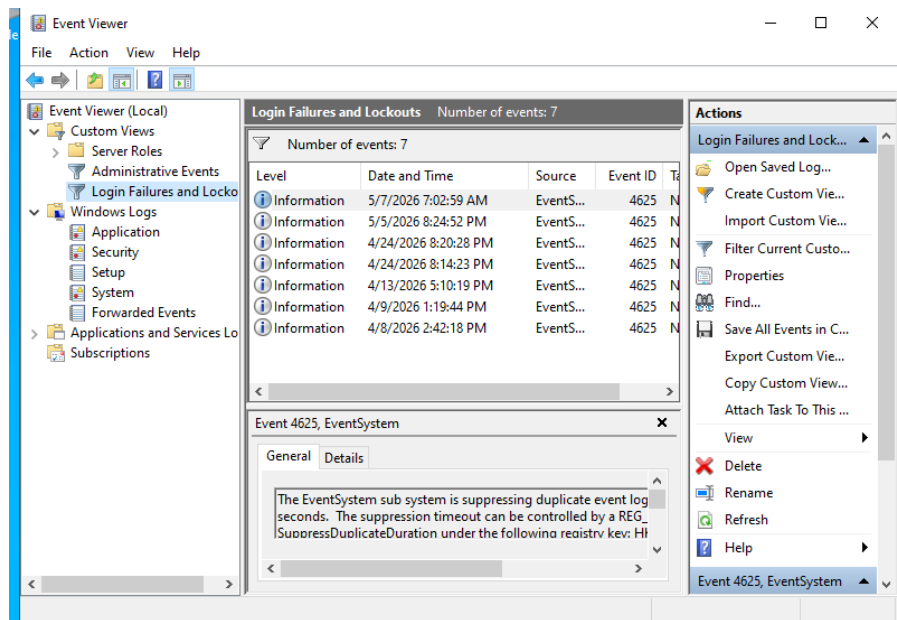
Logged into one of my workstations several times failing it to try to generate a security/setup warning with an event id of 4625 (Failed login)



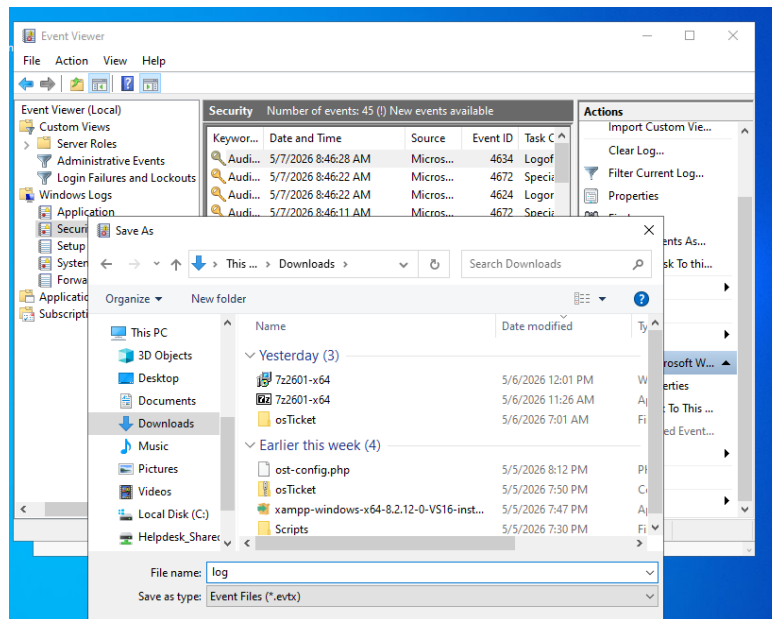
Found login attempt, as well as the previous 5-6 failed logins I attempted



Simulating an error with DNS by temporarily disabling DNS



Created a custom view for both login failures and lockouts with event IDs 4625 and 4740 (Service Failure)



Exported a log as a filter log or as a .evtx in case a senior wants to investigate this alert further.